Final Project Report: Haiku Generation

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Table of Contents

List of figures 3
1. Introduction 4
2. Dataset Review and Preprocessing 5
3. Bigram Model 6
3.1 Implementation 6
3.2 Trade-offs 6
3.3 Experiments and diagnostics 6
3.4 Significance 7
4. GPT-style Models 8
4.1. Version 1: Mini-GPT 8
4.2. Version 2: Small-GPT 8
4.3. Version 3: Lightning-GPT 9
4.4. Version 4: Improved lightning-GPT 9
4.5. Version 5: Word-level-GPT 9
4.6. Version 6: Long word-level-GPT 10
4.7. Version 7: Final-GPT 10
5. Model Comparison 11
6. Limitations and Suggestions 13
7. Conclusion 14
References 15
Appendix 1 16
Appendix 2 22
Appendix 3 23

List of figures

1. Introduction

Haiku is a poem, which typically consists of three lines, 5-7-5 syllable structure (DiMelis & Megregian, 2023).

The problem we raise is whether a language model can learn the structure and constraints of haiku poetry, while producing meaningful flow. Studying language models in such strict scenarios but also favoring creative flow might help us evaluate their abilities better.

We expect the model to be able to capture the general rules, such as syllable or line count; however, it may struggle to produce meaningful flow. Additionally, it may be necessary to state more explicit rules for the model to capture full requirements.

There are several language models for haiku generation; however, we defined some gaps for further research. Firstly, DiMelis & Megregian (2023) focuses on GPT-3, raising the question if other smaller, simpler models would be enough. Secondly, a lot of built models, such as Gerasymiak (2023), used specific prompts, which contributed to more meaningful text but reduced the challenge of independently learned output. Thirdly, Miceli's (2021) model does not impose any correction for the length of the generated haikus, which we believe might lead to many outputs being not actual haikus, thus, we decided to try structure-aware generation. Additionally, we aim to compare sub-level and word-level tokenization effects. Lastly, our study tries to address Bigram vs GPT-2 vs human haiku evaluation differences of computed loss and human preferences.

2. Dataset Review and Preprocessing

The dataset is taken from *Hugging Face* website and contains 49,024 haikus (Figure 1).

Figure 1. *Sample dataset*

	source	text	text_phonemes	keywords	keyword_phonemes	gruen_score	text_punc
6348	bfbarry	Ease all Buddhas. / Resist the artificial. / Dissolve your ego.	ehm|breys aol buw|daxz / rih| jhehkt dhax aar|tax|fih|shaxl / dih| zaalv yaor iy|gow	all buddhas	aol buw|daxz	0.836762	None
33574	haiku_data_1	Lateday, sun. / The cavity of a snag. / Laced with webs.	laet|dey sahn dhax kae|vax|tiy ahv ax snaeg leyst wihdh wehbz	the cavity	dhax kae|vax|tiy	0.437271	Lateday, sun. The cavity of a snag. Laced with webs.
7226	img2poems	The sound of tadpoles. / Singing that late winter day. / Curated my winter blues.	dhax sawnd ahv taed|powlz / sihng|axng dhaxt leyt wihn|ter dey / kyuhrd may wihn|ter bluwz	winter blues	wihn|ter bluwz	0.545987	None
5329	bfbarry	Murmur of the Brook. / Stomachs up the highway roar. / Once your feet get wet.	mer|mer ahv dhax bruhk / swaa| lowz ahp dhax hhay|wey raor / wahns yaor fiyt geht weht	highway roar	hhay|wey raor	0.625580	None
37300	haiku_data_1	New Year's Party. / The divorced couple. / Leaves together.	nuw yihrz paar|tiy dhax dax| vaorst kah|paxl liyvz tax|geh|dher	new year	nuw yihr	0.518931	New Year's Party. The divorced couple. Leaves together.

We see that the dataset contains a few columns but only one ('text') is necessary for us.

It was found that some of the haikus did not follow the 5-7-5 syllable or 3 line rules (Figure 2).

Figure 2. *Syllable count on original data*

```
575 distribution:
is_575
False    30784
True     18240
```

Additionally, some of the haikus contained different notations of line breaks, brackets, spacing, symbols, etc. Therefore, they were removed or fixed to sustain more clarity. We also dropped duplicates and null values when applicable.

After all of the cleaning, we were left with 18,259 real haikus (Figure 3).

Figure 3. Syllable count on the cleaned data

```
575 distribution:
is_575
True    18259
```

From this point we continued to build different models.

3. Bigram Model

The bigram model served as the baseline language model. We evaluated its ability to capture linguistic structure using only word-to-word dependencies. Since Haikus have stylistic constraints, using this model allowed us to see what aspects could be learned without the need for more complex architectures. This also gave us a comparison point for the models that are developed later, checking for any loss of performance. Therefore, the bigram model was a natural first step.

3.1 Implementation

After cleaning the dataset, each haiku was split into tokenized words that were then linked to a numerical ID. The overall vocabulary was restricted to the most frequent terms while adding an <UNK> token for the rarer words so that the model could deal with them. Then, a word-to-word bigram count matrix was built by scanning the whole dataset for pairs of consecutive words. From there, conditional probabilities were estimated using add-α smoothing to avoid zero-probability transitions. The negative log-likelihood over the dataset was used as the main evaluation metric. This approach allowed for a controlled environment where we could directly evaluate the effect of smoothing, vocabulary size, and tokenization choice.

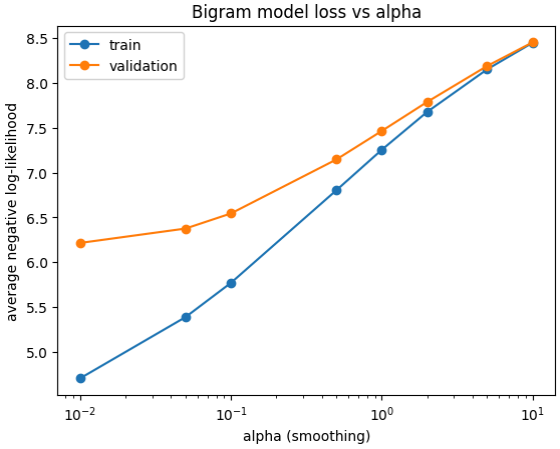
3.2 Trade-offs

The model had a few trade-offs due to its simplicity. It only looked at one preceding word, which limited it to capturing local patterns. Haikus follow a strict 5-7-5 syllable pattern, which requires a model that can learn its broader structure. Word-level tokens made the model interpretable, but it increased sparsity when dealing with a small dataset. Restricting the vocabulary did help, but it limited the variety in word choice even more, increasing the size of the <UNK> category. We also had to balance between too little and too much add-α smoothing, because the former created zero-probability transitions, and the latter flattened probabilities and removed meaningful patterns. These trade-offs confirmed the need for more expressive architectures.

3.3 Experiments and diagnostics

We tested multiple values for α to see their effect on performance. For each α, we computed the negative log-likelihood over the entire dataset, which allowed us to compare how well the model predicted the next word in a haiku. Lower α's seemed to produce sharper probability distributions, but this introduced zero-probability transitions. Higher α's reduced sparsity but flattened the overall structure. Plotting losses across different α's showed a minimum, which was the optimal balance between the two extremes. (Figure 4)

Figure 4. *Negative log-likelihood vs α*




```
alpha=0.01: train_loss=4.7030, val_loss=6.2142
alpha=0.05: train_loss=5.3865, val_loss=6.3749
alpha= 0.1: train_loss=5.7688, val_loss=6.5434
alpha= 0.5: train_loss=6.8023, val_loss=7.1459
alpha= 1.0: train_loss=7.2540, val_loss=7.4643
alpha= 2.0: train_loss=7.6771, val_loss=7.7884
alpha= 5.0: train_loss=8.1528, val_loss=8.1899
alpha=10.0: train_loss=8.4488, val_loss=8.4536
```

We also generated sample text using both the best-performing and worst-performing α to see the differences.

3.4 Significance

The bigram helped shape the direction of the project by starting simple, and serving as a baseline. We saw how much structure could be captured using word dependencies. The experiments using vocabulary size, smoothing, and prediction accuracy revealed the limitations of this approach. Simple probabilistic models fall short and more advanced models are necessary.

4. GPT-style Models

We tried to construct several GPT-2 style models that produce haikus, so we could choose the most accurate one. Our goal was for the model to produce 3-line logical text that follows the 5-7-5 syllable requirements of a haiku, while minimising validation loss.

4.1. Version 1: Mini-GPT

We started off from the simplest-form GPT model with character level tokenization, token and positional embeddings, self attention and feedforward layers, a simple training loop and a generate function. The goal of this model was not to produce any meaningful text, but rather to build a baseline on which we could slowly build a bigger, more complex transformer.

We tried generating some text with this model, but it produced meaningless outputs. The model would most often select a single token and repeat it 200 times. This is because the model is still underdeveloped: we have character level tokenization, no masking of the future values, or any haiku-pattern recognition enforcements in this model yet.

4.2. Version 2: Small-GPT

The following improvements were made for the 2nd model: we made the model slightly bigger (block_size and num_embed 64→128, num_layers 2→4, batch_size 32→64), started tracking training and validation losses, switched to sub-word tokenization, implemented causal masking. We added special character tokens to the text file to distinguish haiku structure (<HAIKU START>, <LB>, <HAIKU END>). We also updated the generate function to break after 3 lines have been generated.

This model rarely manages to generate 3-line text, it is not very haiku-like, it misses the syllable requirements. On the other hand, the text is somewhat readable – the model is producing real words and even sometimes putting them in a logical sequence. This model is clearly able to learn something since training and validation losses were decreasing (Training loss: 1.8824 and Validation loss: 2.2775), showcasing that we can build further on it.

4.3. Version 3: Lightning-GPT

The 3rd model focused on improvements such as increasing the number of heads and training time (2000→3000 iterations), adding dropout (0.2). We also started to use PyTorch Lightning here to have clean training loops. The generate function was updated so it would target 5-7-5 syllables in corresponding lines (used syllapy library). We also updated special character tokens to { | }, so they don't tokenize randomly.

The trained model is trying to recognise the haiku patterns, but it often produces 2 lines and has larger losses (Train Loss: 7.7664, Val Loss: 5.1262) compared to the 2nd version. The outputs themselves are also less coherent compared to version 2. This is likely happening due to strong regularization, overcomplexity, and the amount of data that the model is learning from – it is failing to capture patterns and predict the next token in a logical way, it still requires some fine-tuning.

4.4. Version 4: Improved lightning-GPT

Version 4 mostly focused on updating hyperparameters to reduce loss and get more accurate, logical haikus. Mostly we were trying to reduce some hyperparameters since we might be trying to implement too complex of a model on too small of the data. We reduced block_size to 64, num_heads to 4, num_layers to 2, dropout to 0.15. Also tried improving the generate function to better reinforce the syllable structure.

This model managed to reduce loss more compared to version 3 (Train Loss: 4.5900, Val Loss: 4.3221). But we still fail to generate exactly 3 lines each time indicating something might still be wrong within the generate functions restrictions. The outputs still lack meaning or structure. This indicates that our model is undertrained and failing to capture meaning, small training data and limited resources restrict us when building similar models, thus simplification of the code is needed.

4.5. Version 5: Word-level-GPT

In version 5 we decided to try word-level tokenization since the syllable counts might be better captured when the model is ensured to generate real words. We updated hyperparameters once again. Decreased the block_size to 32 this time, since the haiku doesn't need much context to work on. Decreased the learning rate for smoother convergence. Increased num_embed, num_heads, num_layers so the model can learn more complex parameters.

Even though the validation loss was a bit higher compared to model 4 and training loss (Train Loss: 4.7050, Val Loss: 4.6986), we sometimes managed to generate 3-line text that follows the 5-7-5 syllable structure. But the text still does not have much meaning.

4.6. Version 6: Long word-level-GPT

In an attempt to improve the meaning of our haikus, we decided to train on a bigger haiku dataset where not all haikus follow the required syllable structure. Our data increased by an additional ~22 thousand examples. The model itself stayed the same as version 5.

This time we got bigger losses than in the 5th version (Train Loss: 5.1754, Val Loss: 5.2183), but the haikus have more logical meaning, they are usually 3-lines and match the syllable requirements. The higher validation loss indicates that we might start overfitting a little, which is expected since we're using word-level tokenization and quite a small dropout.

4.7. Version 7: Final-GPT

This is our final attempt to improve our model, we keep using the longer dataset to have better meaning. Another issue – the model keeps producing a 4th line if the syllable count requirement is not meant, we update our generate function to never produce a 4th line (even if the syllable counts are a bit off). We also increased the training iterations and regularization to restrict overfitting, we (dropout 0.3 and weight decay 0.1).

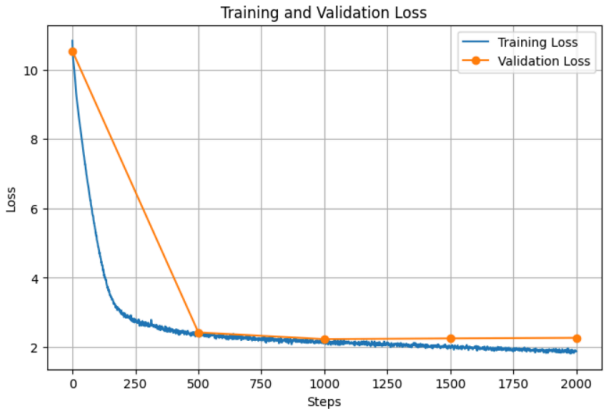
This model produces outputs that are still not perfect syllable-wise, and loss could still be lower. But we finally managed to cut the model off after generating 3 lines of text and it is way more likely to produce the right number of syllables in each row regardless of not having the lowest loss out of all models. We also managed to prevent overfitting by implementing stronger regularization. Overall, this model produces the most haiku-like text out of all of our models, thus we chose it as our best and final version.

5. Model Comparison

For comparison, we calculated each model's losses and conducted human evaluation for better understanding of what is considered a good haiku.

As we can see, the best training loss and validation loss was in the model, which did not produce correct structure or meaningful haikus (Figure 5) (Appendix 1 presents all of the other loss graphs and examples).

Figure 5. *Small-GPT Loss and Examples*




```
Haiku 1:

agack mickers
 The rain and my thoughts of the

---------------------------------------

Haiku 2:

ense the morning
 I just got into my life

---------------------------------------

Haiku 3:

aken person
 Ill get some happy that I

---------------------------------------
```

Worse losses were present in all other models. The second best loss is seen in the Improved-Lightning-GPT. Additionally, almost all of the better models, especially those who finally reach proper structure and some meaning, presented overfitting as validation loss exceeded the training loss.

The best training and validation loss ratio (Appendix 2) was present in the bigram model with alpha=10.0, whose loss is big but it doesn't overfit or underfit (because it's really simple and barely learns anything). Also, in word-level, long word-level and final GPTs. Word-level GPT presented the lowest loss out of all of the mentioned models; however, it often lacked any meaning in the text, with final GPT sometimes presenting better outputs (note: it also used word-level tokens).

Additionally, a survey was conducted (results in Appendix 3) with 21 respondents to have a more fair and unbiased evaluation of the haikus. We presented them with three examples of each type of haiku: Real, Bigram with alpha=0.01, Small-GPT, Lightning-GPT, Improved-Lightning-GPT, Word-level GPT, Final-GPT. They were evaluated on three categories: fluency and coherence, syllable count, creativity. Fluency and coherence winner was the real haiku, with next being the word-level and final-GPT haikus. Correct syllable count winner was the real haikus; however, final-GPT model came into not that far second, with third place being 2 times lower word-level GPT. Finally, for the creativity factor answers distributed similarly as in the first category. Thus, by human evaluation, the best models that

we built were the word-level and final-GPTs, suggesting the value of word-level tokens on human preference.

Therefore, even if the models did not seem to be the best according to the losses, human evaluation confirmed that more realistic haikus are the word-level token ones. However, they were definitely not as good as the real haikus. This can be explained by the fact that it is hard for the model to capture the meaning of each haiku, as they do not show any significant similarities, except for the strict structure.

6. Limitations and Suggestions

During our modelling, we faced limitations and trade-offs:

- The trade-off between *input text accuracy and the amount of input text*.The dataset was contaminated with other text (not all haikus), so, at first, we decreased the usable information from the dataset significantly. Additionally, even some of the haiku-looking data contained the needed structure but some lacked poetic meaning, and it was hard to distinguish between them, which might have led to worse results of our models (e.g. curse words in the outputs). However, only having accurate structure haikus did not lead us to the best results and having more random text helped the model to generalize better.
- The trade-off between *model complexity and training time.* For example, version 3 and 4 took forever to load as they were the most complicated ones.
- Sometimes we also witnessed that there is a trade-off between the *complexity of the model and the quality of output,* as the most complex or the simplest models (versions 3 and 4) were not the best and we had to find the optimal middle.
- The trade-off between *loss reduction and output accuracy.* The simplest models in the beginning had the lowest loss, but did not resemble haikus, while complex models had higher loss due to overfitting but had better outputs. We balanced it out by introducing increased regularization and word-level tokenization for simplicity.
- The trade-off between *correct output lines and correct syllable counts.* For the model it was hard to learn both and sometimes it started to overshoot due to very strict rules, thus, we settled for correct line counts.

Suggestions for future research:

- To evaluate models with syllable accuracy metrics (we did conduct human evaluation; however, machine evaluation might also be useful as humans sometimes tend to be incorrect in counting the syllables as well).
- To try out other mixes of the models (for example, mixing some components of model 4, which had quite low loss, with model 7, which had quite good output).

7. Conclusion

Using a dataset with different text we built multiple models (bigram and GPT-2 style), without using prompts or any initialization words but with structure-aware generation, only focusing model learnability and output on itself. The best models (averaging quantitative and qualitative evaluation) were the word-level-GPT and final-GPT, which did not have the smallest losses out of all models but the best outputs (especially the latter). This shows that loss alone is not a reliable indicator, especially of poetic quality. Additionally, we have learned that, in this case, word-level tokenization was a better choice for a proper output, even if it showed signs of overfitting more (however, when trained on more data (final-GPT), it decreased).

In summary, our initial hypothesis was confirmed and even though the models did learn the structure of the haiku (3 lines and 5-7-5 syllables) with the help of integrated structure-aware generation, it still made mistakes and did not reach the level of poetism as humans would. Thus, even complex models of this research did not show the overall best learning abilities.

References

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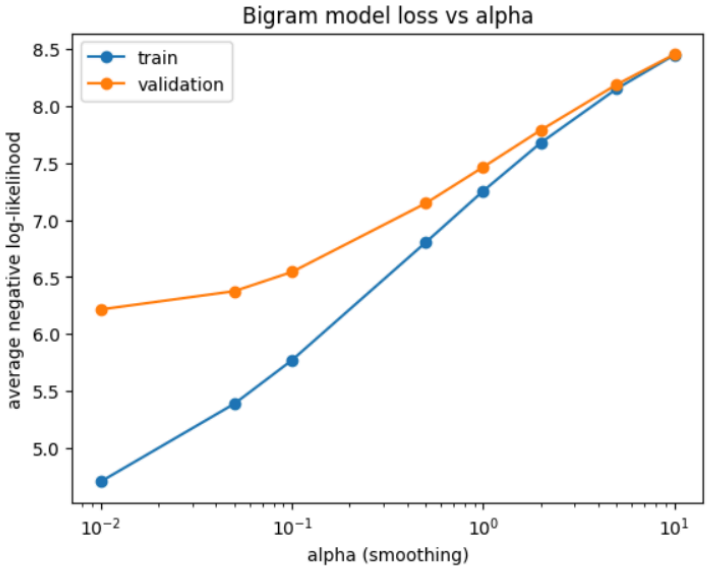
Appendix 1

Bigram model losses:




```
Best alpha based on validation loss: 0.01
```

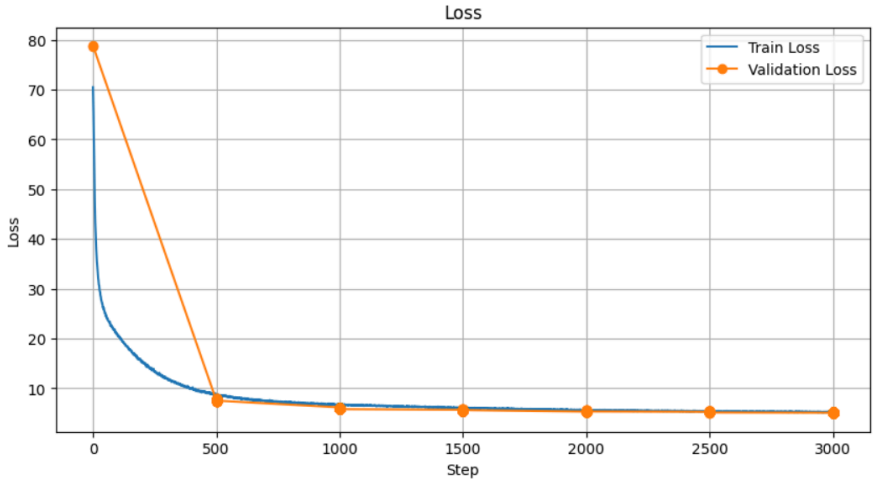
Bigram model examples:

```
Generated text with BEST alpha = 0.01
- suffer retard regular nicki wax obama ascended place under the holidays outfits denny pearl listening doggo downpours nope collection photos
- mob plow cried along chugging pendulum spurs spouse israel its beautiful edgy already getting afghanistan outlets nipples chop niw spoiler
- windmills attempting delightfully span leftists magnolia coo ushua thunderstorm boaling plugging myself stench vikings winning poorly balls selling softening michael
- steaming entire whish pricks skies existence stack handmade voice monster muller purpose gossamer avoid monochromy unicorns provoking feating press rang
- reserved lap relationships randle chivalry allah coward baggage chosen become a heavy conclusion spies dimming backtrack dime built freeze distracts

Generated text with BAD alpha = 10.0
- fund supple disappoint resonated hidels true original snacks discussions flows replaced loss fights wino tempest lil bastards chore struggle anchor
- marches dry foul daffodil raleigh westerns sir robert toy depleted echoes cursing popper flutter sized creep courtney armed listeners interviews
- grandma sword worships advil bank spoiler embrace center husky ten gucci lab whooshing tom northern womans flickers giddy tomatoes server
- branches pendulum voice stupidity cigarette manager scum firefighters bye bass fourth bite baltimore reviews effortless petition independence minerals pancake status
- hypocrite strewn consciousness row great covered costume starbucks breaths frees devoid firewood thanos piqueing rounding notes nation comforter crashy snack
```

Mini-GPT example (presented an empty cell as a haiku):

```
step 0 | loss 4.1323
step 200 | loss 2.6106
step 400 | loss 2.5057
step 600 | loss 1.7719
step 800 | loss 0.2668
step 1000 | loss 0.1353
step 1200 | loss 0.0740
step 1400 | loss 0.0620
step 1600 | loss 0.0558
step 1800 | loss 0.0522
step 2000 | loss 0.0499
```

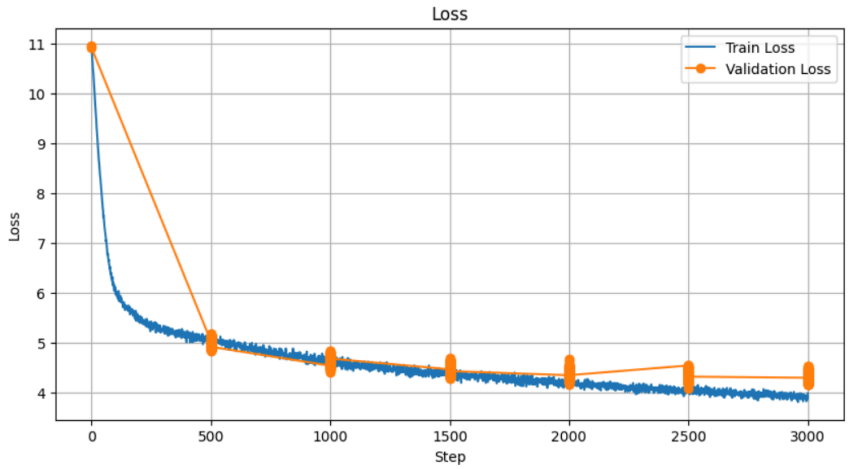
Lightning-GPT loss:



Lightning-GPT examples:

```
Haiku 1:

your my
T me for the

---------------------------------------


Haiku 2:

the do so
I have of the is

---------------------------------------


Haiku 3:

in a
You to you not

---------------------------------------
```

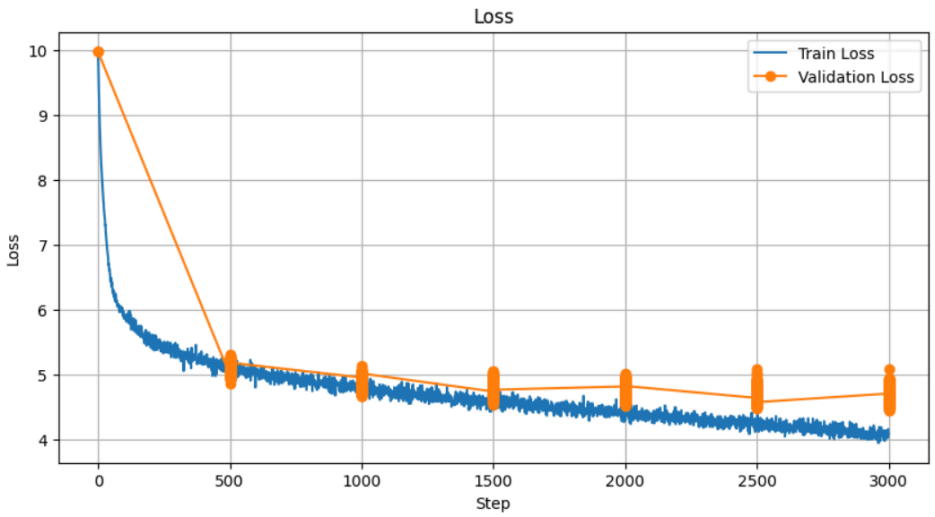
Improved-Lightning-GPT loss:



Improved-Lightning-GPT examples:

```
Haiku 1:

iced on
We would be nice after the
You are nice

Merry
To

----------------------------------------


Haiku 2:

old dream
I need to be here and
For a lot

I always
I

----------------------------------------


Haiku 3:

iced
I love my hair

I hate being
It was the same that
Its

----------------------------------------
```

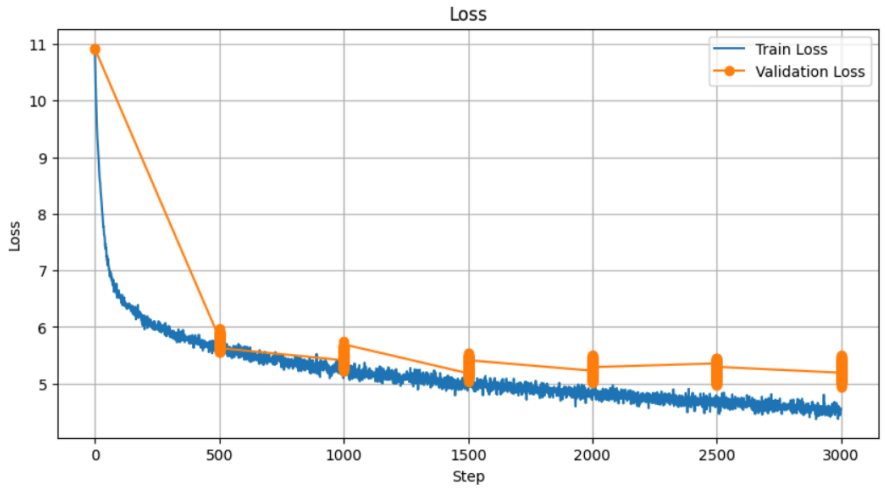
Word-level GPT loss:



Word-level GPT examples:

```
Haiku 1:

I honestly cant
 Stop doing the first thing to
 Say it in the night

----------------------------------------


Haiku 2:

Your baby told me
 Going to work on my heart
 In love is everyone
 For

----------------------------------------


Haiku 3:

The moon is hard
 When hes a wonderful
 Thing is a good day
 And

----------------------------------------
```

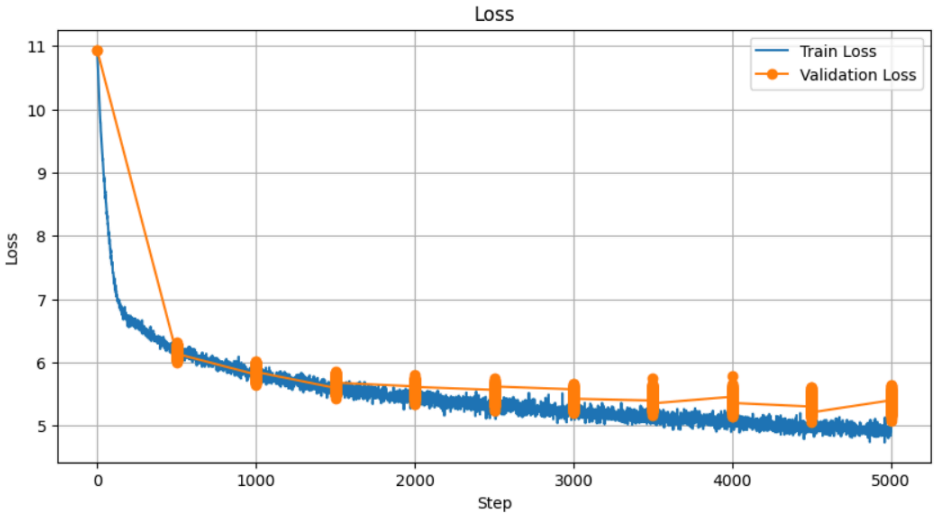
Word-level-GPT-on-Bigger-Data loss:



Word-level-GPT-on-Bigger-Data examples:

```
Haiku 1:

I don't wanna talk.
 You know I'm not sure I.
 Be an easy to
 How

----------------------------------------


Haiku 2:

Winter morning.
 A cat of the light.
 In the garden wind.
 The

----------------------------------------


Haiku 3:

What do I am.
 To do you can't be.
 Don't listen to do.
 To

----------------------------------------
```

Final-GPT loss:



Final-GPT examples:

```
Haiku 3:

I just wanna be.
 The best I have no one for
 To be a man.   You

----------------------------------------


Haiku 4:

A new year
 The street of a year.
 In a little morning.

----------------------------------------


Haiku 5:

I know one don't.
 Get is a nice I don't.
 You are not not on.

----------------------------------------
```

Appendix 2

The losses in numbers:

Bigram:	alpha=0.01: train_loss=4.7030, val_loss=6.2142
Bigram:	alpha=10.0: train_loss=8.4488, val_loss=8.4536
	Small GPT: Training loss: 1.8824 and Validation loss: 2.2775
	Lightning-GPT: Train Loss: 7.7664, Val Loss: 5.1262
	Improved lightning-GPT: Train Loss: 4.5900, Val Loss: 4.3221
	Word-level GPT: Train Loss: 4.7050, Val Loss: 4.6986
	Long word-level GPT: Train Loss: 5.1754, Val Loss: 5.2183
	Final-GPT: Train Loss: 5.5071, Val Loss: 5.3796

The ratios of losses (validation loss/training loss):

Bigram	Ratios (val/train)
alpha=0.01	1.321326813
alpha=10.0	1.000556292
Small GPT	1.209891628
Lightning-GPT	0.660048414
Improved lightning-GPT	0.941633987
Word-level GPT	0.998639745
Long word-level GPT	1.008289214
Final GPT	0.976848069

Appendix 3

Fluency and coherence category answers and calculations:

Haiku:																					Average answer:	Average of three same type haikus:	
tim tea is it durant yet moustache crackers juices sentence gain chaps buffalo chicken restrain dictate brutality tomorrows dropping predicted	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.031746032	Bigram
reborn disaster hallucinate throat mindfully liquor daisy unproductive breathing kinship linear outburst snatch length khalid stumbled bathe laguna ton abomination	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
dips casting question next to care about my stretch tread claus yield offspring lame mix educate lyrics professor packing stuffed	1	1	2	1	1	1	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1.095238095		
Delicate savage.You'll never hold the cinder.But still you will burn.	5	5	5	5	5	5	5	5	4	5	5	5	5	5	4	5	5	5	3	5	2	4.666666667	Real
Steamy, mist rising.Rocks receiving downward crash.As the jungle weeps.	5	5	5	5	5	5	5	5	5	2	5	5	3	5	3	3	5	5	5	3	5	4.476190476	
A splash and a cry.Words pulled from the riverside.Dryed in the hot sun.	5	5	5	5	4	5	5	5	2	5	5	2	5	3	3	5	5	5	3	5	4	4.333333333	
rastination But that will be no more	1	1	3	1	2	2	1	1	2	1	1	1	1	1	1	2	1	1	1	1	1	1.26984127	Small GPT
ense the morning I just got into my life	1	1	3	1	1	2	2	1	2	1	1	4	1	1	1	1	1	1	1	1	1	1.380952381	
s me, baby's DayA big words, my baby's	1	1	3	1	1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.142857143	
your my T me for the	1	1	1	1	1	1	1	1	1	1	1	1	1	1		1	1	1	1	1	1	1	1.031746032	Lightning GPT
the do so I have of the is	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		
in a You to you not	1	1	3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.095238095	
iced on We would be nice after the You are nice Merry To	1	1	4	1	1	2	1	1	1	1	1	2	2	1	2	1	2	1	2	1	1	1.428571429	1.523809524	Improved lightning GPT
old dream I need to be here and For a lot I always I	1	1	4	1	2	2	1	1	1	1	1	2	2	1	2	1	2	1	2	1	1	1.476190476	
iced I love my hair I hate being It was the same that Its	1	1	4	1	1	3	1	1	1	1	1	5	2	2	1	2	1	2	1	2	1	1.666666667	
I could be happy And I miss my parents allTo see you again	3	3	3	1	1	3	4	1	3	1	5	5	3	3	1	3	1	3	1	3	2	2.523809524	2.650793651	Word-level GPT
The problem is the Life is my baby And you have to say My	3	3	3	1	1	3	3	1	2	1	1	3	3	3	1	3	1	3	1	3	2	2.142857143	
I hate my boyfriendBut I am so happyAnd I am not supposed	5	3	5	1	2	5	3	3	3	1	5	5	4	4	4	5	1	3	1	3	3	3.285714286	
Don't wanna be here. Be one are not all my. To get you doing?	3	3	4	1	2	3	2	3	2	1	5	2	3	2	2	4	1	4	1	4	1	2.523809524	2.485714286	Final GPT
The sun is not know. Who say you can keep trying. Be a good day. It	3	3	3	1	2	2	2	3	2	1	5	1	3	2	3	3	1	3	1	3	2	2.333333333	
I need someone to. Go back to you wanna be? What an adult What	3	3	3	1	3	2	3	3		1	5	1	3	2	3	4	1	4	1	4	2	2.6	

Syllable count category answers and calculations:

Haiku:																					NO:	YES:	Average of three no's	Average of three yes's:	
tim tea is it durant yet moustache crackers juices sentence gain chaps buffalo chicken restrain dictate brutality tomorrows dropping predicted	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0			Bigram
reborn disaster hallucinate throat mindfully liquor daisy unproductive breathing kinship linear outburst snatch length khalid stumbled bathe laguna to	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0	21	0	
dips casting question next to care about my stretch tread claus yield offspring lame mix educate lyrics professor packing stuffed	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0			
Delicate savage.You'll never hold the cinder.But still you will burn.	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	0	21			Real
Steamy, mist rising.Rocks receiving downward crash.As the jungle weeps.	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	0	21	0.333333333	20.66666667	
A splash and a cry.Words pulled from the riverside.Dryed in the hot sun.	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	1	20			
rastination But that will be no more	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0			Small GPT
ense the morning I just got into my life	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0	20.66666667	0.333333333	
s me, baby's DayA big words, my baby's	NO	NO	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	20	1			
your my T me for the	NO	NO	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	20	1			Lighting GPT
the do so I have of the is	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0	20.66666667	0.333333333	
in a You to you not	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0			
iced on We would be nice after the You are nice Merry To	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0			Improved lightning GPT
old dream I need to be here and For a lot I always I	NO	NO	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	20	1	20.33333333	0.666666667	
iced I love my hair I hate being It was the same that Its	NO	NO	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	20	1			
I could be happy And I miss my parents allTo see you again	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	YES	YES	YES	NO	YES	2	19			Word-level GPT
The problem is the Life is my baby And you have to say My	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	21	0	13	8	
I hate my boyfriendBut I am so happyAnd I am not supposed	NO	NO	YES	NO	NO	NO	YES	NO	NO	NO	NO	YES	NO	NO	NO	YES	NO	NO	NO	YES	16	5			
Don't wanna be here. Be one are not all my. To get you doing?	YES	YES	NO	NO	NO	NO	NO	YES	NO	NO	YES	YES	NO	YES	YES	YES	NO	NO	NO	YES	12	9			Final GPT
The sun is not know. Who say you can keep trying. Be a good day. It	YES	YES	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	1	20	4.666666667	16.33333333	
I need someone to. Go back to you wanna be? What an adult What	YES	YES	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	1	20			

Creativity category answers and calculations:

Haikus:																					Average:	Average of three same type haikus:	
tim tea is it durant yet moustache crackers juices sentence gain chaps buffalo chicken restrain dictate brutality tomorrows dropping predicted	1	3	5	1	1	2	2	1	5	1	3	3	2	1	3	2	2	1	2	1	5	2.2381	Bigram
reborn disaster hallucinate throat mindfully liquor daisy unproductive breathing kinship linear outburst snatch length khalid stumbled bathe laguna ton abominatio	1	3	5	1	2	2	3	1	5	1	3	3	2	1	3	3	2	1	2	1	5	2.38095	2.301587302
dips casting question next to care about my stretch tread claus yield offspring lame mix educate lyrics professor packing stuffed	1	3	5	1	1	2	3	1	5	1	3	3	2	1	3	2	2	1	2	1	5	2.28571	
Delicate savage.You'll never hold the cinder.But still you will burn.	3	4	5	5	4	4	5	5	5	5	3	5	5	3	4	4	5	5	5	3	5	4.38095	Real
Steamy, mist rising.Rocks receiving downward crash.As the jungle weeps.	3	5	5	5	5	4	5	4	5	5	3	5	5	4	4	5	5	5	5	3	5	4.52381	4.476190476
A splash and a cry.Words pulled from the riverside.Dryed in the hot sun.	3	5	5	5	5	4	5	5	5	5	3	5	5	4	3	5	5	5	5	3	5	4.52381	
rastination But that will be no more	1	1	5	1	1	1	2	2	5	1	1	3	1	2	2	1	1	1	1	1	5	1.85714	Small GPT
ense the morning I just got into my life	1	1	5	1	1	1	2	1	5	1	1	4	1	2	1	1	1	1	1	1	5	1.80952	1.777777778
s me, baby's DayA big words, my baby's	2	1	5	1	1	1	2	1	5	1	1	2	1	2	2	1	1	1	1	1	2	1.66667	
your my T me for the	1	1	5	1	1	1	1	1	5	1	1	1	1	1	1	1	1	1	1	1	1	1.38095	Lightning GPT
the do so I have of the is	1	1	5	1	1	1	1	1	5	1	1	1	1	1	1	1	1	1	1	1	1	1.38095	1.380952381
in a You to you not	1	1	5	1	1	1	1	1	5	1	1	1	1	1	1	1	1	1	1	1	1	1.38095	
iced on We would be nice after the You are nice Merry To	2	1	5	1	1	1	4	1	5	1	2	2	2	1	2	2	2	1	2	1	4	2.04762	Improved lightning GPT
old dream I need to be here and For a lot I always I	2	1	5	1	1	1	2	1	5	1	2	1	2	1	2	1	2	1	2	1	4	1.85714	2.095238095
iced I love my hair I hate being It was the same that Its	3	1	5	1	1	1	4	1	5	1	2	5	2	3	3	2	2	1	2	1	4	2.38095	
I could be happy And I miss my parents allTo see you again	4	2	5	1	4	3	5	2	5	1	3	5	3	3	4	3	3	3	3	2	5	3.28571	Word-level GPT
The problem is the Life is my baby And you have to say My	4	3	5	1	3	3	2	2	5	1	3	3	3	2	4	3	3	3	3	1	5	2.95238	3.492063492
I hate my boyfriendBut I am so happyAnd I am not supposed	5	5	5	1	4	5	4	3	5	5	4	5	5	5	5	5	5	4	3	4	2	5	4.2381		
Don't wanna be here. Be one are not all my. To get you doing?	3	4	5	1	2	3	3	3	5	5	3	2	3	4	4	4	4	3	4	2	5	3.42857	Final GPT
The sun is not know. Who say you can keep trying. Be a good day. It	3	3	5	1	2	4	4	2	5	5	3	4	4	4	4	4	3	3	3	2	5	3.47619	3.412698413
I need someone to. Go back to you wanna be? What an adult What	3	3	5	1	2	2	3	1	5	5	3	2	4	5	4	4	4	3	4	2	5	3.33333	